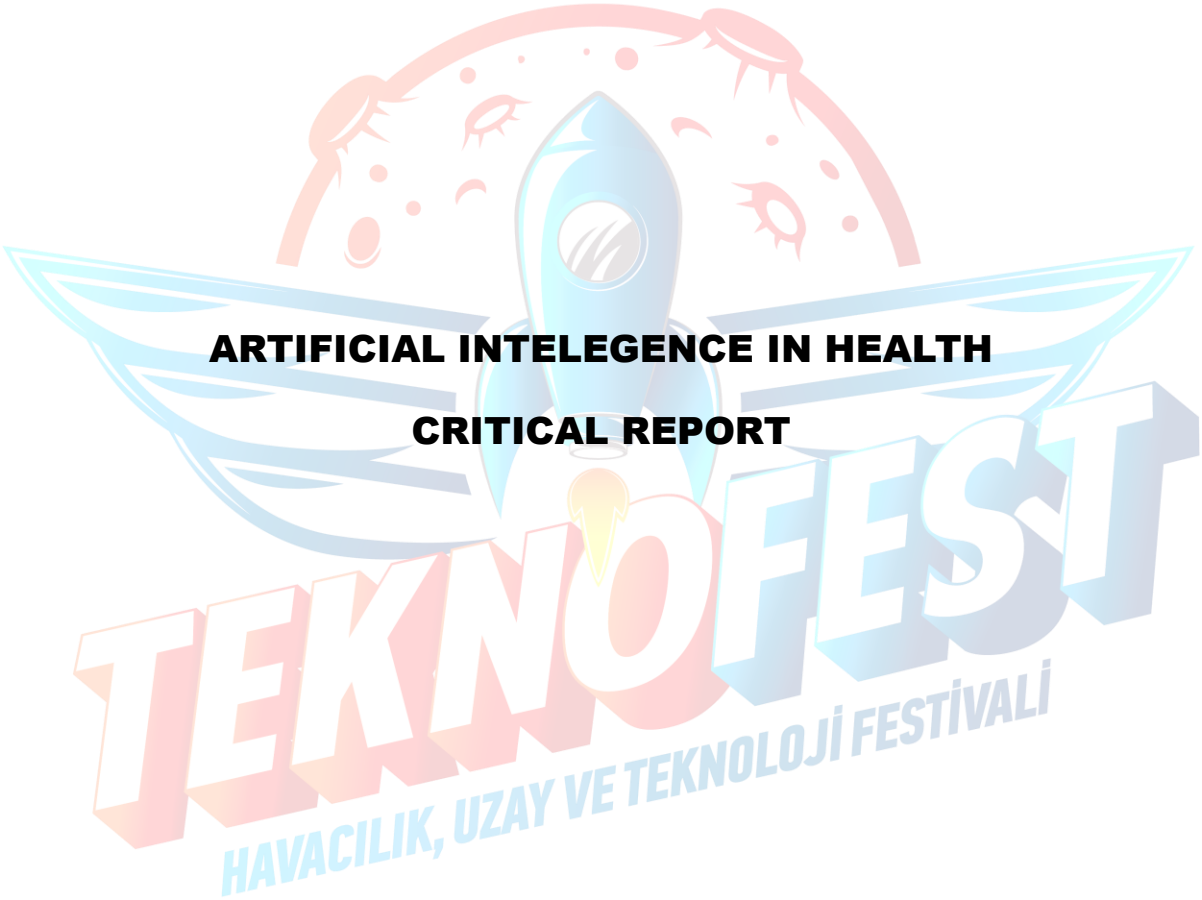
# TEKNOFEST

# AEROSPACE AND TECHNOLOGY FESTIVAL




## ARTIFICIAL INTELEGENCE IN HEALTH

## CRITICAL REPORT

**TEAM'S NAME**

**CHATGPTREE**

**APLICATION ID**

**3321623**

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## 1. TEAM CHART

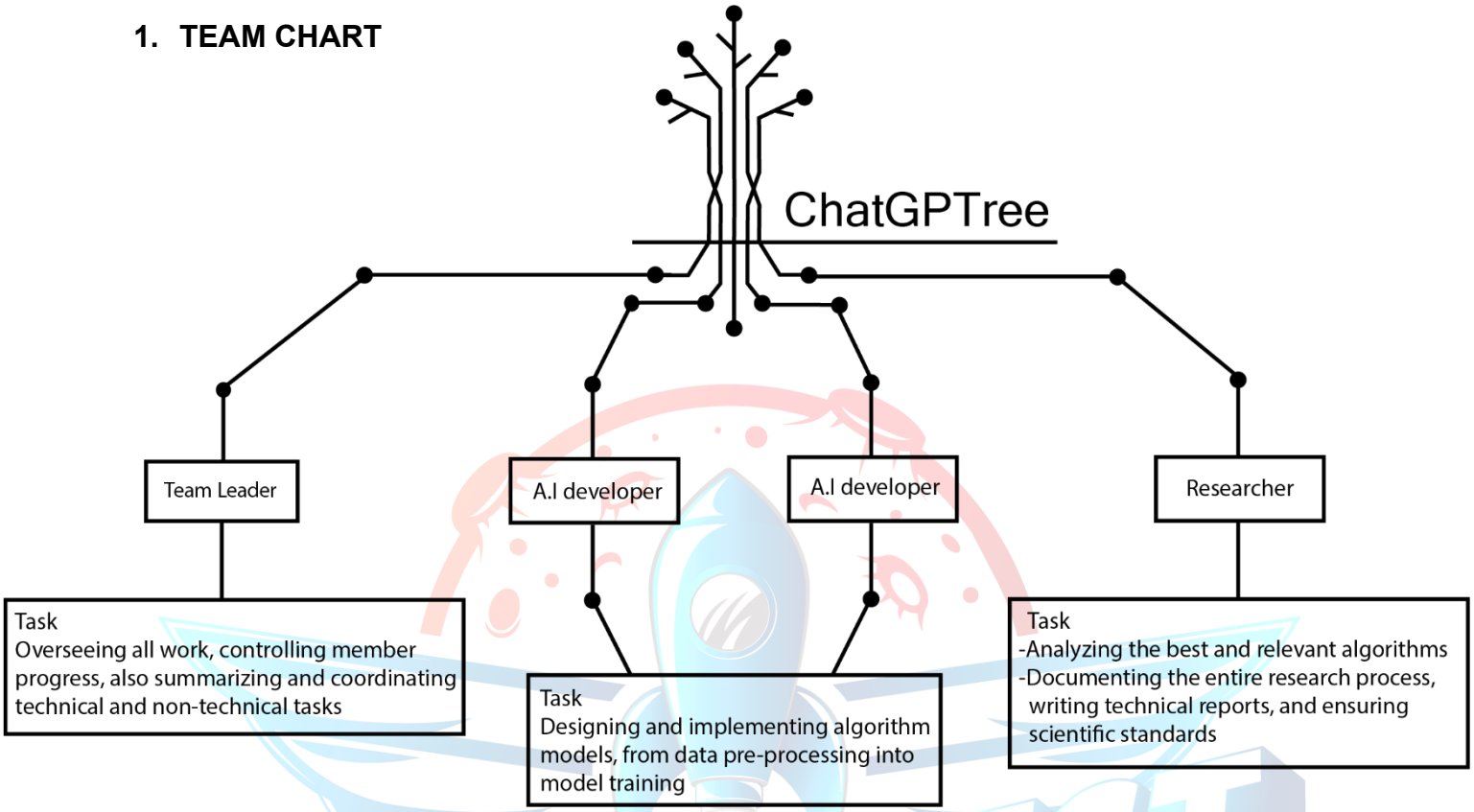



***Figure 1.*** *Team Chart*

## 2. SUMMARY OF INTERNATIONAL PAPER

### 2.1. Cardiothoracic Ratio Calculation from Chest X-rays

Cardiothoracic Ratio (CTR) is the ratio of heart size to chest cavity on a chest X-ray, used to detect cardiomegaly (enlarged heart) and monitor conditions such as heart disease. The transverse cardiac diameter is measured as the maximum horizontal distance between the rightmost and leftmost borders of the cardiac silhouette, where the right border is defined by the lateral border of the right atrium A and the left border by the apex of the left ventricle B. The transverse thoracic diameter is then measured as the widest internal horizontal distance between the inner borders of the ribs at the level of the right hemidiaphragmatic dome C, typically corresponding to the ninth or tenth posterior rib. CTR is obtained by dividing the transverse cardiac diameter by the transverse thoracic diameter and multiplying the quotient by 100 to express the result as a percentage. CTR exceeding 50% in adults indicates cardiomegaly, although pediatric thresholds vary with age. [1][9]

$$CTR = \frac{A + B}{C} \times 100$$

### 2.2. CNN Regression Method

One of the key strengths of CNNs lies in their minimal need for manual engineering, making them highly adaptable across domains. This versatility has enabled CNNs to become specialized in real-world applications, e.g. computer vision, speech processing, medical imaging, and natural language processing. Moreover, they are applied in autonomous systems to support navigation in robotics and vehicles. [2]

However, CNNs are not limited only to complex problems and tasks such as classification or segmentation, they are also powerful to solve one of the classical problems, regression tasks, where the goal is to predict continuous values. The example application of CNN for regression task is the estimation of RUL (Remaining Useful Life) of a system or component from sensor data. Beyond that, a CNN regression approach is implemented to achieve real-time 2-D/3-D registration. [3][5]

In summary, we understand that regression tasks are not trivial problems for CNNs rather it is worth notetaking that solving regression task, which is one of the classical problems of machine learning, with modern implementations through CNNs show significant advancements in algorithmic design and performance. Recognizing this, we choose to adopt a regression-based approach in our model in predicting anatomical coordinates for CTR computation.

## 3. DATASETS AND ALGORITHM USED

### 3.1. Datasets

In this study, we used one dataset to validate the generalization of the model. In the dataset split, we applied the 80:20 ratio referring to the study of recent which proved that the splitting of 80% training data and 20% validation data is optimal to balance the training needs of the model while maintaining sufficient data for performance evaluation without overfitting. We use the dataset provided by Teknofest. This dataset has 102 raw images along with their coordinates of chest anatomies. [6]

### 3.2. Algorithm Used

MobileNetV2 is a compact convolutional neural network designed specifically for mobile and embedded devices, where computational power and memory are limited. Building on the success of MobileNetV1, V2 introduces two key innovations inverted residual blocks and linear bottlenecks that allow it to remain extremely lightweight while achieving strong performance on tasks like image classification, object detection, semantic segmentation and even regression. [7]

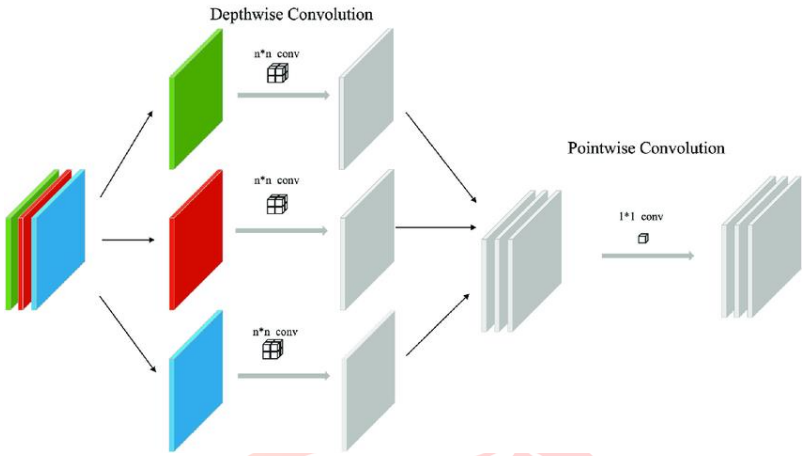



***Figure 2.*** *Mechanism of Depthwise Separable convolutions*

The mechanism of depthwise separable convolutions that MobileNetV2 uses, is to split the work into two steps, that is filtering each channel separately (depthwise convolution), then combining them with a quick 1×1 convolution to cut computations by ~8–9× without much loss in accuracy. [7]

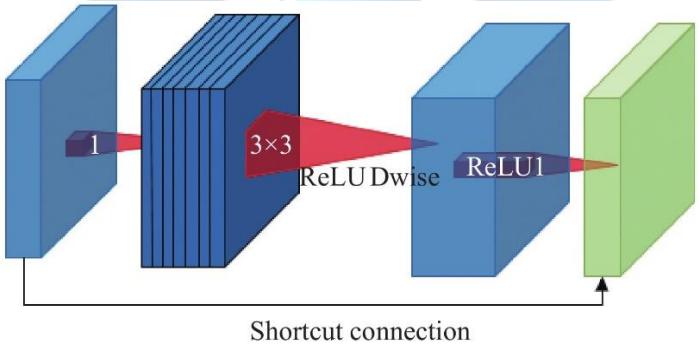



***Figure 3.*** *Inverted Residual Blocks*

The inverted residual blocks in MobileNetV2 first expand the number of channels, then apply an efficient depthwise convolution channel-wise, and finally project back to a smaller channel count. This "wide → narrow" sequence, paired with skip connections, helps maintain information flow and gradient stability throughout the network without increasing computational cost. [7]

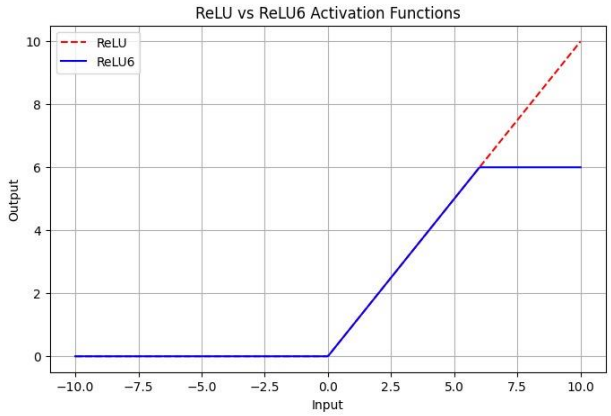



***Figure 4.*** *ReLU and ReLU6 Comparison*

Most networks use a ReLU activation after each layer, which works well in high-dimensional spaces; but in tight, narrow layers (bottlenecks), it can destroy information by setting many values to zero. MobileNetV2 addresses this by using a linear bottleneck, skipping nonlinear activations like ReLU in the final 1×1 projection layer to better preserve important details. Combined with depthwise separable convolutions, inverted residuals, linear bottlenecks, and ReLU6, these design choices make MobileNetV2 both extremely efficient and surprisingly accurate on mobile and embedded devices. [7]

## 4. RESULT AND ANALYSIS

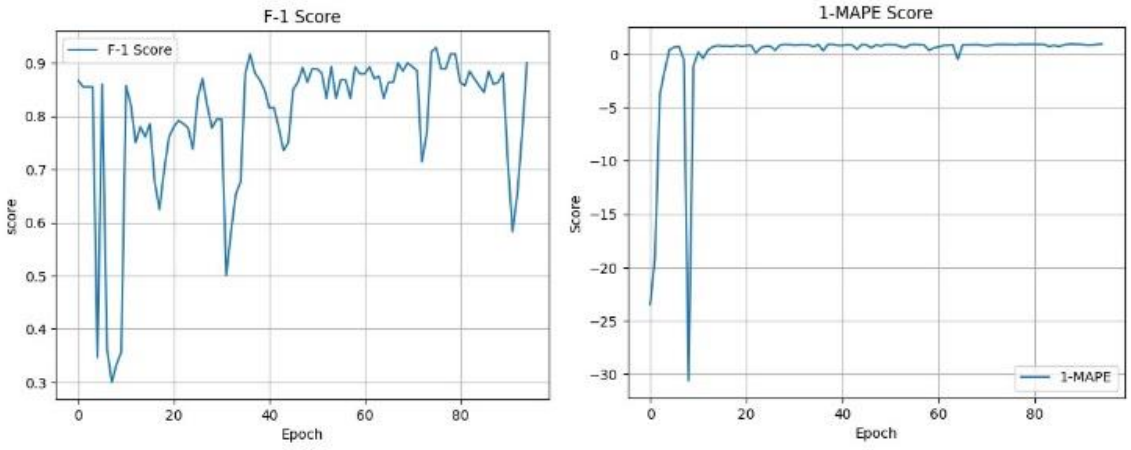



***Figure 5.*** *Evaluation Score Graphic*

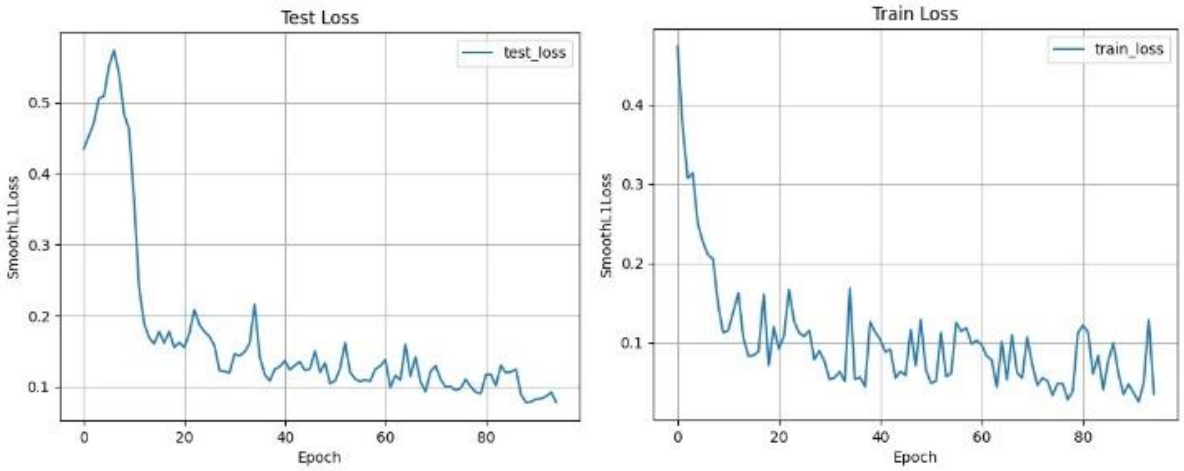



***Figure 6.*** *Test and Train Loss Graphic*

The graphs presented above illustrate the model's progress, showing the loss values and the evaluation scores achieved during training. Before executing the training loop, we tune the value of hyperparameters (such as batch size, learning rate, and epoch), and determine the best loss function and optimizer to ensure optimal performance.

Although we applied transfer learning to our model, the dataset consists of medical images, which typically require more complex and deeper feature extraction. To give the model sufficient time to adapt and learn from these features, we trained it for 95 epochs. Initially, we used a learning rate of 1e-5, but the model showed low accuracy and slow convergence. After experimenting, we found that increasing the learning rate to 1e-4 a commonly used value led to better performance. We settled on a moderate batch size of 8 which is the most efficient based on our experiments results.

**Formula of Smooth L1 Loss**

$$L(x) = \begin{cases} 0.5(x-y)^2, & \text{if } |x-y| < 1 \\ |x-y| - 0.5, & \text{otherwise} \end{cases}$$

where $x$ is the predicted bounding box coordinate and $y$ is the ground truth.

***Figure 7.*** *Formula of Smooth L1 Loss*

In the context of regression method based on CNN, we implement the SmoothL1Loss to guide the prediction of anatomical coordinates which serve the basis for predicting CTR. We selected SmoothL1Loss function because the robustness to outliers where anatomical annotation noise may occur. SmoothL1Loss function is the benefit combination of L1 loss and L2 loss which in turn makes SmoothL1Loss less

sensitive to extreme values; therefore, maintaining stability and smoothness, and reducing the tendency of overfitting. [8]

$$F1\ Score = 2 \times \frac{Precision \times Recal}{Precision + Recal}$$

$$MAPE = \frac{\sum \left| \frac{Actual - Forecast}{Actual} \right| \times 100\%}{Sample\ Size}$$

After the training phase, we apply F1 score and 1-MAPE for the evaluation metrics, F1 score for cardiomegaly classification and 1-MAPE for the CTR calculation accuracy. F1 score is the harmonic mean of Precision and Recall and is widely used in binary classification problems. It matters because it balances between false positive and false negative. A high F1 score indicates both high precision (few false positive) and high recall (few false negative). 1-MAPE converts error measure of MAPE, which measures the average absolute difference between predicted and actual values (expressed as percentage), into an accuracy-like measure. [8]

We experimented with three widely used optimizers for our project: SGD, Adam, and AdamW. Eventually, AdamW optimizer outperforms the other optimizer. AdamW is the modification of Adam algorithm where weight decay regularization decoupled from the gradient-based parameter updates. This feature can lead to a better generalization and better weight regularization because it applies consistent penalization to all parameters regardless of gradient history. [4]

## 5. ORIGINALITY

### 5.1. Image Preprocessing

The major problem we dealt in the given dataset is about scale. All the images are different in dimension, and this issue can cause a significant problem for CNN to be able to solve pattern in the images. To address this problem, we rescale all the images into the ideal size that common CNN model could receive, 224 by 224 pixels. After rescaling the images' dimension, we need to rescale each of the coordinate proportional to the rescaled images because the coordinates still correspond with the original images. Moreover, in the further step these coordinates take role as the ground truth for the predicted coordinates that will be generated by the CNN model.

The data augmentation pipeline combines geometric transformations and photometric adjustments techniques to enhance model robustness and generalization. For the image normalization step, we applied the mean and standard deviation values commonly used for ImageNet which are based on its three-color channels (RGB), mean

= [0.485, 0.456, 0.406] and std = [0.229, 0.224, 0.225]. The pipeline balances realism and diversity to improve model performance across clinical and geometric variability.

### 5.1.1. Geometric transformations

Geometric transformations are mathematical operations that modify the position, orientation and size, by applying coordinate mapping functions.

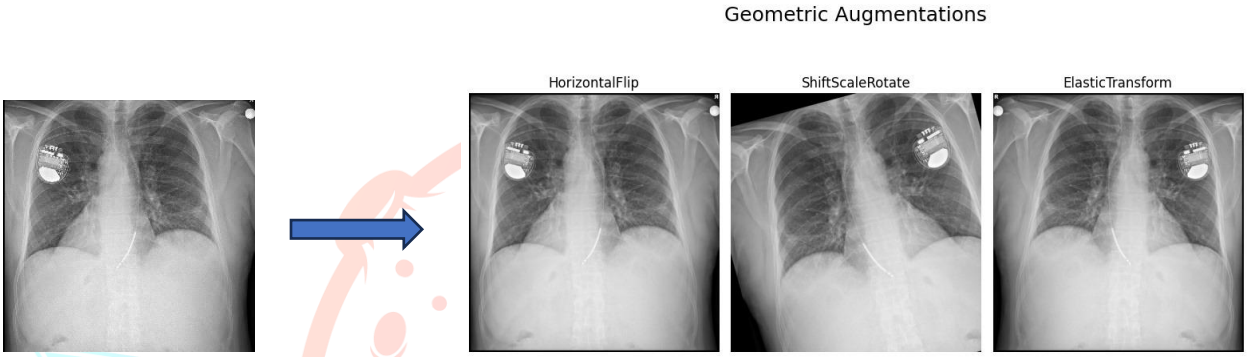



***Figure 8.*** *From Original to Geometric Augmented Chest X-Ray*

### 5.1.2. Photometric Adjustments

Photometric adjustments are image processing operations that modify pixel intensity values such as brigthness, contrast, clahe, gauss noise and gaussian blur to enhance features like cardiac borders in X-rays without altering spatial geometry.

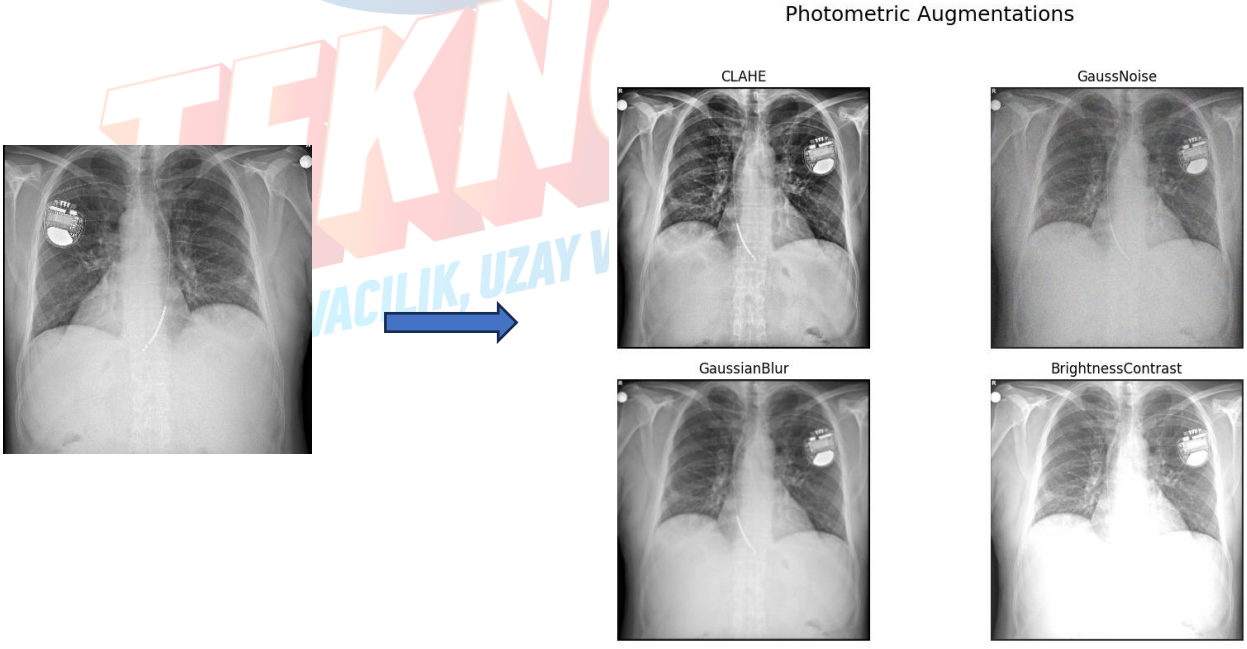



***Figure 9.*** *From Original to Photometric Augmented X-Ray Chest*

### 5.1.3. Regularization

Regularization in image preprocessing is a technique used to clean up and improve image quality. It works by adding simple rules to the image data to help reduce unwanted noise and to make the image more clear and consistent. These rules might include making the image look smoother, less cluttered, or keeping important edges sharp. This technique ensures the model focus on the important parts of the image, while ignoring unnecessary details. For instance, we implemented a common regularization technique called coarse dropout, which randomly removes rectangular region from the image. This encourages the model to become more robust by learning to recognize key features even when parts of the image are missing.

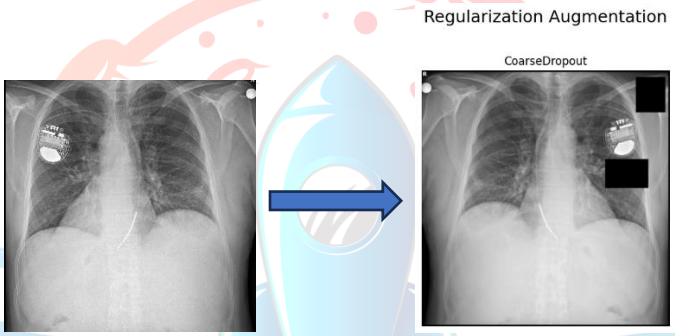



***Figure 10.*** *From Original Dataset Chest X-Ray Image to Regularization Augmented*

## 5.2. Ground Truth Preprocessing

Our model is designed to output 12 values, representing the (x, y) pairs of 6 coordinates of chest anatomies. However, these predictions still correspond with the scale of normalized images, meaning they cannot be directly compared with original (unnormalized) coordinates. To address this, we need to normalize the true coordinates by computing their mean and standard deviation in the training loop so that we do not mess up with the value of true coordinates in the initial step. This step ensures that both true coordinates and predicted coordinates operate on the same scale, allowing us to calculate the loss.

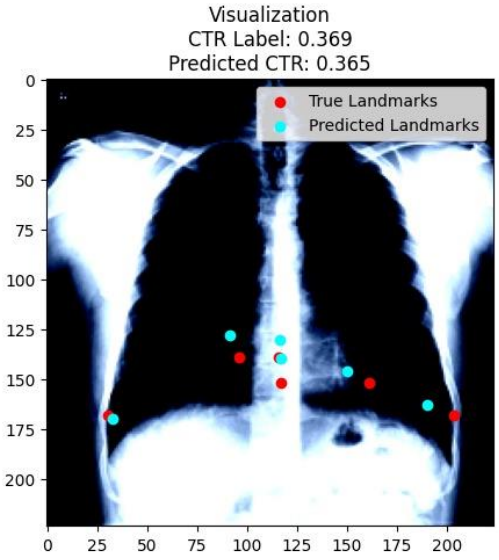



***Figure 11.*** *Visualization of Predicted and True Landmarks*

Once the model is trained, we reverse the normalization on the predicted coordinates using the previously computed mean and standard deviation. This step restores the predictions to their original spatial scale, making them interpretable. To verify the model's performance, we progressively visualize the predicted and true coordinates, the normalized image, and both the true and predicted CTR labels, as illustrated in Figure 11. From these restored coordinates, we can compute the predicted CTR and use it to evaluate the model using both MAPE and F1 score, ensuring accuracy in both regression and classification.

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